Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. The learning rate was set to 0.0003 with the Adam optimizer. Each experiment was repeated 3 times with different random seeds.

2 Results

Figure 1 shows the test accuracy by dataset.

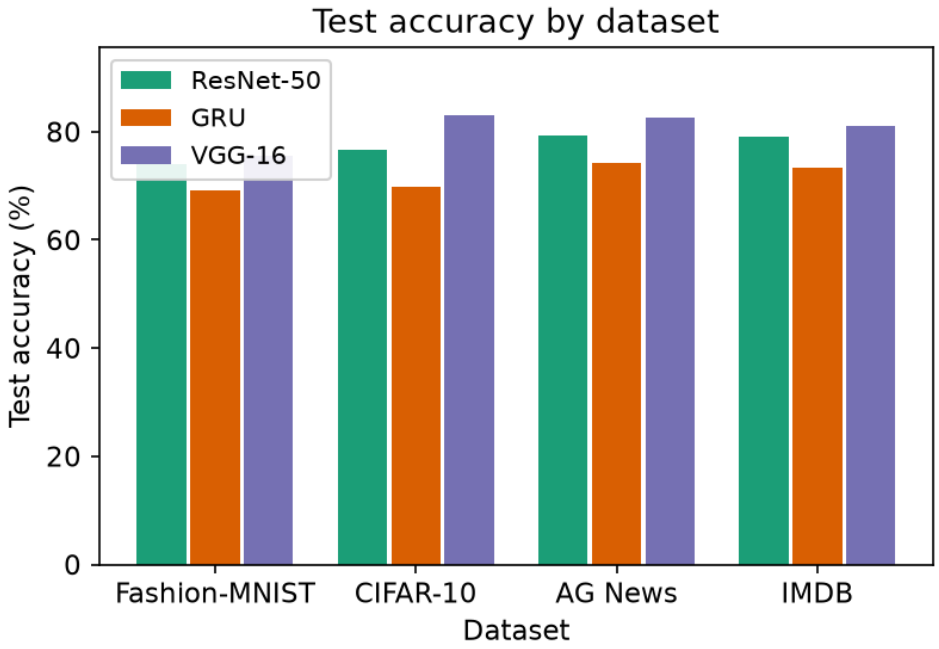


Figure 1: Test accuracy by dataset for the models in this experiment.

On Fashion-MNIST, VGG-16 is about 6 points better than GRU. VGG-16 achieves 84.4% test accuracy on AG News. As shown in Figure 1, GRU achieves about 70% test accuracy on IMDB. ResNet-50 outperforms GRU by 5 percentage points on AG News (Figure 1).

Figure 2 shows the test accuracy by dataset.

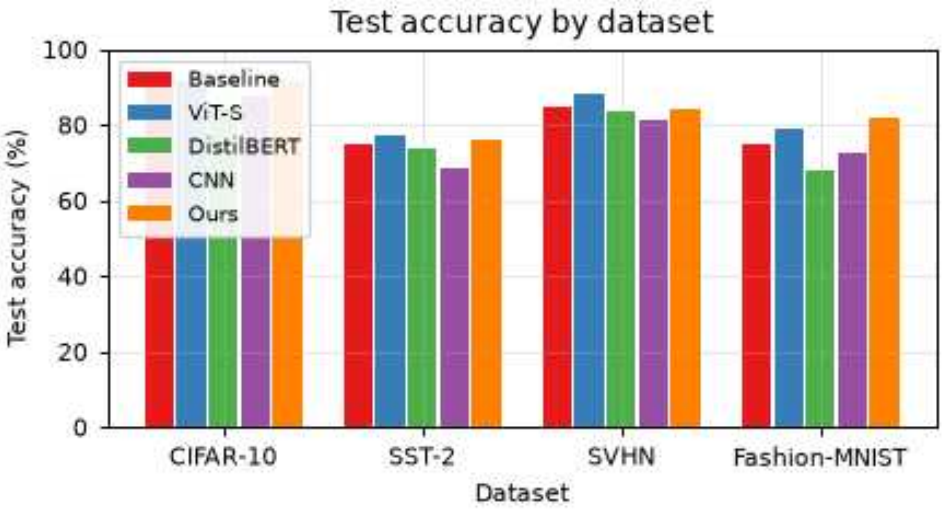


Figure 2: Test accuracy by dataset for the models in this experiment.

CNN reaches over 85% test accuracy on CIFAR-10. Baseline reaches 83.9% test accuracy on CIFAR-10 (Figure 2). As shown in Figure 2, on SST-2, ViT-S is 7.2 points better than Baseline.

Figure 3 shows the accuracy by dataset.

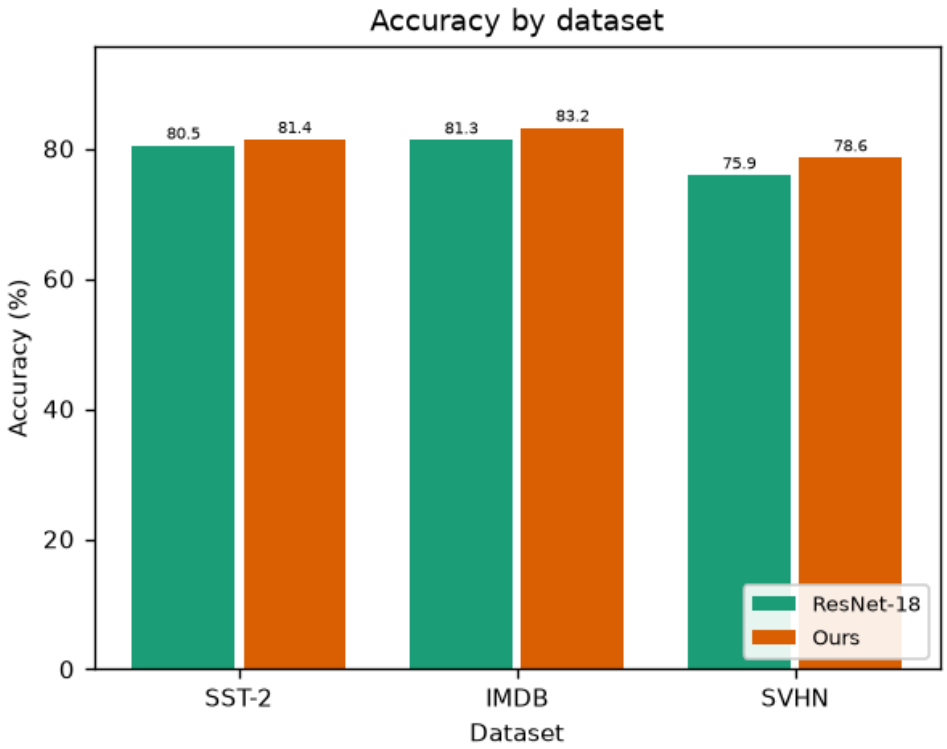


Figure 3: Accuracy by dataset for the models in this experiment.

On SST-2, Ours obtains a accuracy of 79.5%. As shown in Figure 3, Ours achieves 62.3% accuracy on SST-2.

3 Discussion

SVM reaches 93.3% accuracy on SST-2. Overall, the results are in line with what we expected from the literature.